

Revised Generation Interconnection System Impact Study Report

For Queue Project Q18-1126

Ibm Cottle Tap 100 kV
December 21, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1126. The Interconnection Customer proposes a 144.7 MW solar photovoltaic facility interconnecting at the Ibm Cottle Tap 100 kV substation in the Indiana area. The request was received on December 21, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 448 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,564,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Ibm Cottle Tp Substation - Bear Garden 100 kV	Ibm Cottle Tap Substation - Arden Hills 100 kV	116.0	108	100.0
Muskogee - Montpelier Electric Generating Sta 500 kV	Lonetree - Sabanera 500 kV	103.1	4,673	12.1
24Th & Gardendale - Sidney A Murray Jr Hydroelectric 230 kV	Richland - Dekalb 500 kV line	100.5	860	8.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ibm Cottle Tp Substation - Bear Garden 100 kV	Rebuild line with higher-capacity conductor	\$4,109,000	100% (\$4,109,000)
Muskogee - Montpelier Electric Generating Sta 500 kV	Reconductor line	\$50,562,000	0% (\$243,000)
Sidney A Murray Jr Hydroelectric - 24Th & Gardendale 230 kV	Reconductor line	\$14,135,000	2% (\$212,000)
	Total allocated to project		\$4,564,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.